DATA ITEM DESCRIPTION
Title: POD FATIGUE AND DAMAGE TOLERANCE METHODOLOGY REPORT

Number: DI-SESS-82264
AMSC Number: N10029
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20190516
Limitation: N/A
GIDEP Applicable: N/A
Project Number: SESS-2019-017

Use/relationship: The Pod Fatigue and Damage Tolerance Methodology Report specifies the fundamental engineering assumptions and analysis methodologies that will be used to substantiate the pod's design with respect to fatigue and damage tolerance. The data will be used to develop the Stress, Fatigue, and Fracture Analysis Report.

This Data Item Description (DID) contains the format and content for data resulting from the work tasks described by the contract statement of work.

Requirements:

1. Reference documents. None.

2. Format. Contractor format acceptable.

3. Content. The Pod Fatigue and Damage Tolerance Methodology Report shall contain the following:

3.1 Describe the methodology used to perform fatigue and damage tolerance analyses of the pod structure, including joints, fasteners, and fittings.

3.2 Describe the fatigue life and damage tolerance criteria for the pod.

3.3 Describe the spectrum generation process and the method by which the repeated loads defined by the master event spectrum will be filtered and utilized in the fatigue and damage tolerance analyses.

3.4 Describe the methodology to be used to determine or develop fatigue allowables and the criteria for using strain-life or stress-life crack initiation curves, and crack growth rate curves.

3.5 Describe any design policies or guidelines used in lieu of detailed fatigue analyses, such as fastener de-rating policies used in lieu of detailed fastener fatigue analysis, or any proposed allowable stress cut-off levels that are used in lieu of structure detail fatigue analysis.

3.6 Describe the methodology to be used to account for stress concentrations due to geometry effects. Describe hole geometry effects, loaded fastener hole effects, hole filling effects, fastener and hole fit effects, complex loading and combined stress concentration effects, notches effects, fillet effects, multiaxial stress states, shallow stress gradient effects, shim and gap effects, and any other geometry effect in the designed pod structure that differs from a simple geometry stress concentration found in published and approved handbooks or proprietary vendor test data.

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3.7 Describe the methodology used to account for material, processing, and environmental effects in the fatigue and damage tolerance analyses including fatigue life enhancement processes or methods.

3.8 Describe any planned deviations from standard aerospace industry accepted fatigue and damage tolerance analysis practices. Describe the method to verify the fatigue and damage tolerance analyses for the pod.

End of DI-SESS-82264

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